Data Mining Analysis of the Statlog (German Credit Data) Dataset

Using the `ucimlrepo` Python package (UCI Machine Learning Repository, dataset id 144)

Report date: 05 October 2026

Executive summary

- **Dataset.** 1,000 loan applicants, 20 attributes (7 numeric, 13 categorical), binary target *good* (700) vs *bad* (300) credit risk. There are no missing values and no duplicate rows. The classes are imbalanced 70:30.
- **Strongest single driver: checking-account status.** The bad-credit rate is 49% for an overdrawn account (< 0 DM) and 12% for applicants with no checking account at this bank (Cramer's V = 0.35). Credit history, loan duration, savings and property follow.
- **Longer, larger loans are riskier.** The median duration is 24 months for bad loans vs 18 months for good loans. Loans over 36 months combined with savings under 100 DM default 72% of the time (association rule, lift 2.4).
- **Predictive models.** Seven classifiers were compared with stratified 10-fold cross-validation. Random Forest, SVM and logistic regression lead with a ROC-AUC of about 0.79-0.80. Accuracy at the default 0.5 threshold is about 75-77%, only modestly above the 70% obtained by labelling everyone "good".
- **The cost matrix changes the decision rule.** Because a missed bad customer costs 5 and a rejected good customer costs 1, the cost-optimal threshold is P(bad) > 1/6, not 0.5. This cuts cross-validated cost by roughly 40-50% for most models (e.g. SVM (RBF): 893 to 508). It also beats the trivial "reject everyone" policy (cost 700).
- **Fairness.** Age, sex/marital status and foreign-worker status are linked to the target. Any operational model should be audited for disparate impact before use (Section 9).

1. Methodology and data acquisition with ucimlrepo

The study follows the CRISP-DM data-mining process: business understanding, then data acquisition and understanding, pre-processing, modelling (classification, clustering, association rules), evaluation and interpretation. The data are loaded with the `ucimlrepo` package, following the usage instructions in its README (version 0.0.7):

```
pip install -U ucimlrepo

from ucimlrepo import fetch_ucirepo, list_available_datasets

# import dataset (Statlog German Credit Data = id 144)
statlog_german_credit_data = fetch_ucirepo(id=144)
# alternatively: fetch_ucirepo(name='Statlog (German Credit Data)')

# access data
X = statlog_german_credit_data.data.features   # 20 feature columns
y = statlog_german_credit_data.data.targets    # 'class' column (1 = good, 2 = bad)

# access metadata and variable information
print(statlog_german_credit_data.metadata.uci_id)
print(statlog_german_credit_data.metadata.num_instances)
print(statlog_german_credit_data.metadata.additional_info.summary)
print(statlog_german_credit_data.variables)
```

Note on data source. The analysis code calls `fetch_ucirepo(id=144)` first. The environment that produced this report had no access to archive.ics.uci.edu, so the call raised *ConnectionError*. The code then rebuilt the same `dotdict` object (`data.features`, `data.targets`, `data.original`, `metadata`, `variables`) from the supplied *statlog+german+credit+data.zip* (file *german.data*). It used the same column names (Attribute1-Attribute20, class) as the UCI API. The data are therefore identical, and with internet access the script uses the API automatically.

Business objective

A bank wants to decide, from the information on a loan application, whether the applicant is a good or a bad credit risk. The dataset includes a **cost matrix**: classifying a bad customer as good costs 5, and classifying a good customer as bad costs 1. The data-mining goals are to (i) describe which applicant characteristics are associated with bad credit, (ii) build and evaluate predictive models under this cost structure, and (iii) find natural customer segments and frequent risk patterns.

2. Dataset metadata and variables

Field	Value
uci_id / name	144 / Statlog (German Credit Data)
repository_url	https://archive.ics.uci.edu/dataset/144/statlog+german+credit+data
dataset_doi	10.24432/C5NC77
creators	Hans Hofmann
task / characteristics	Classification / Multivariate
num_instances / num_features	1000 / 20
feature_types	Categorical, Integer
target_col	class
has_missing_values	no
summary	This dataset classifies people described by a set of attributes as good or bad credit risks. Comes in two formats (one all numeric). Also comes with a cost matrix.

The `variables` table (name, role, type, demographic, description, units) is shown below with the short names used in the rest of this report.

UCI name	Short name	Role	Type	Demographic	Description	Units
Attribute1	checking_status	Feature	Categorical		Status of existing checking account	
Attribute2	duration	Feature	Integer		Duration	months
Attribute3	credit_history	Feature	Categorical		Credit history	

UCI name	Short name	Role	Type	Demographic	Description	Units
Attribute4	purpose	Feature	Categorical		Purpose	
Attribute5	credit_amount	Feature	Integer		Credit amount	DM
Attribute6	savings	Feature	Categorical		Savings account/bonds	
Attribute7	employment	Feature	Categorical	Other	Present employment since	
Attribute8	installment_rate	Feature	Integer		Installment rate in percentage of disposable income	
Attribute9	personal_status_sex	Feature	Categorical	Marital Status	Personal status and sex	
Attribute10	other_debtors	Feature	Categorical		Other debtors / guarantors	
Attribute11	residence_since	Feature	Integer		Present residence since	
Attribute12	property	Feature	Categorical		Property	
Attribute13	age	Feature	Integer	Age	Age	years
Attribute14	other_installment_plans	Feature	Categorical		Other installment plans	
Attribute15	housing	Feature	Categorical	Other	Housing	
Attribute16	existing_credits	Feature	Integer		Number of existing credits at this bank	
Attribute17	job	Feature	Categorical	Occupation	Job	
Attribute18	num_dependents	Feature	Integer		Number of people being liable to provide maintenance for	
Attribute19	telephone	Feature	Binary		Telephone	
Attribute20	foreign_worker	Feature	Binary	Other	Foreign worker	
class	target	Target	Binary		1 = Good, 2 = Bad	

3. Data quality and descriptive statistics

Shape: 1000 rows x 20 features. **Missing values:** 0. **Duplicate rows:** 0. All categorical attributes use the A-codes defined in *german.doc*. They were decoded to readable labels for the tables and charts. No cleaning or imputation was required.

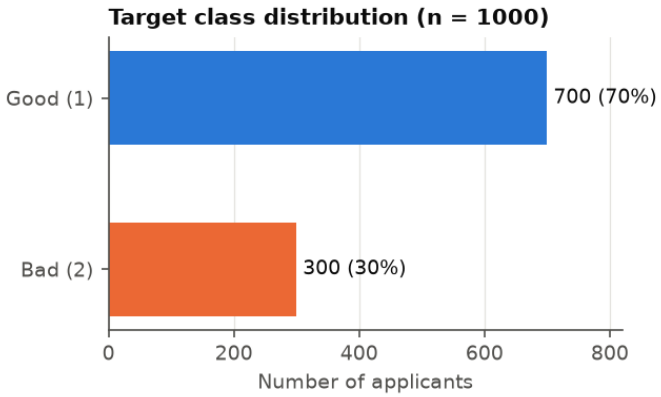


Figure 1. Target distribution. A majority-class classifier would score 70% accuracy, so accuracy alone is a weak metric here.

Variable	Mean	Std	Min	Q1	Median	Q3	Max	Skew
duration	20.9	12.06	4.0	12.0	18.0	24.0	72.0	1.09
credit_amount	3271.26	2822.74	250.0	1365.5	2319.5	3972.25	18424.0	1.95
installment_rate	2.97	1.12	1.0	2.0	3.0	4.0	4.0	-0.53
residence_since	2.84	1.1	1.0	2.0	3.0	4.0	4.0	-0.27
age	35.55	11.38	19.0	27.0	33.0	42.0	75.0	1.02
existing_credits	1.41	0.58	1.0	1.0	1.0	2.0	4.0	1.27
num_dependents	1.16	0.36	1.0	1.0	1.0	1.0	2.0	1.91

Credit amount (median 2,320 DM, max 18,424 DM) and duration are strongly right-skewed. Age ranges from 19 to 75 (median 33). The ordinal attributes take only a few distinct values: installment rate, residence and number of credits range from 1 to 4, and dependents from 1 to 2.

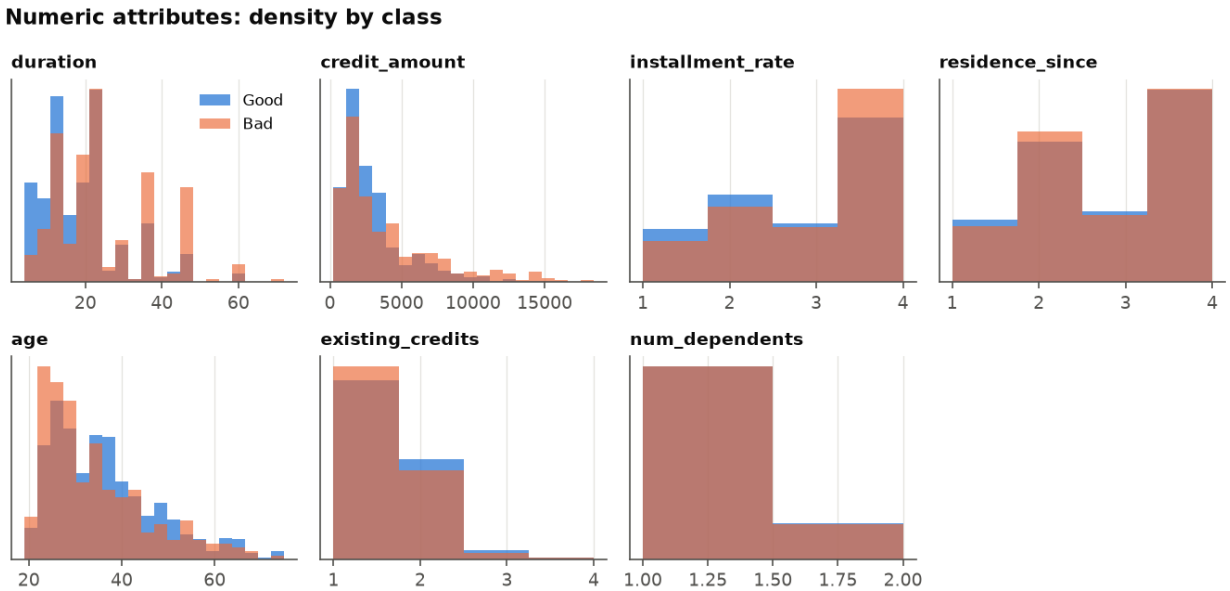


Figure 2. Density of each numeric attribute for good (blue) and bad (orange) applicants. Bad loans shift towards longer durations, higher amounts and younger ages.

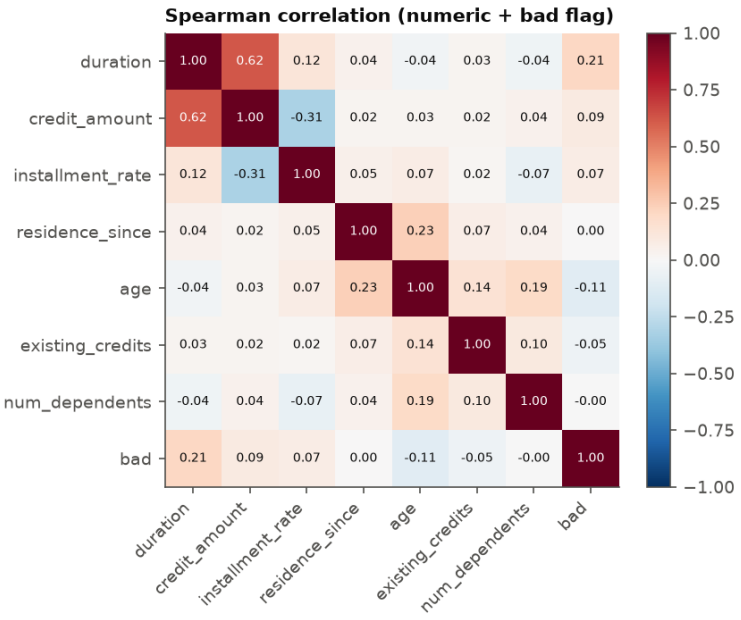


Figure 3. Spearman rank correlations. Duration and credit amount are strongly correlated (rho = 0.62). Duration has the largest correlation with the bad flag.

4. Which attributes are associated with bad credit?

Each attribute was tested against the target. Categorical attributes used a chi-square test of independence, with Cramer's V as the effect size. Numeric attributes used a Mann-Whitney U test, with the rank-biserial correlation as the effect size.

#	Attribute	Test	Statistic	p-value	Effect size	Median good / bad
1	checking_status	chi-square	123.72	< 0.001	0.352	
2	credit_history	chi-square	61.69	< 0.001	0.248	
3	duration	Mann-Whitney U	77995.5	< 0.001	0.257	18 / 24
4	savings	chi-square	36.1	< 0.001	0.19	
5	property	chi-square	23.72	< 0.001	0.154	
6	housing	chi-square	18.2	< 0.001	0.135	
7	purpose	chi-square	33.36	< 0.001	0.183	
8	age	Mann-Whitney U	119833.0	< 0.001	0.141	34 / 31
9	employment	chi-square	18.37	0.001	0.136	
10	other_installment_plans	chi-square	12.84	0.002	0.113	
11	credit_amount	Mann-Whitney U	93480.0	0.006	0.11	2244 / 2574.5
12	foreign_worker	chi-square	5.82	0.016	0.076	
13	installment_rate	Mann-Whitney U	95889.5	0.020	0.087	3 / 4
14	personal_status_sex	chi-square	9.61	0.022	0.098	
15	other_debtors	chi-square	6.65	0.036	0.082	
16	existing_credits	Mann-Whitney U	110272.0	0.135	0.05	1 / 1
17	telephone	chi-square	1.17	0.279	0.034	
18	job	chi-square	1.89	0.597	0.043	
19	num_dependents	Mann-Whitney U	105250.0	0.924	0.002	1 / 1
20	residence_since	Mann-Whitney U	104680.5	0.936	0.003	3 / 3

15 of 20 attributes are significant at the 5% level. **Not significant:** existing_credits, telephone, job, num_dependents, residence_since. These carry little standalone information about credit risk.

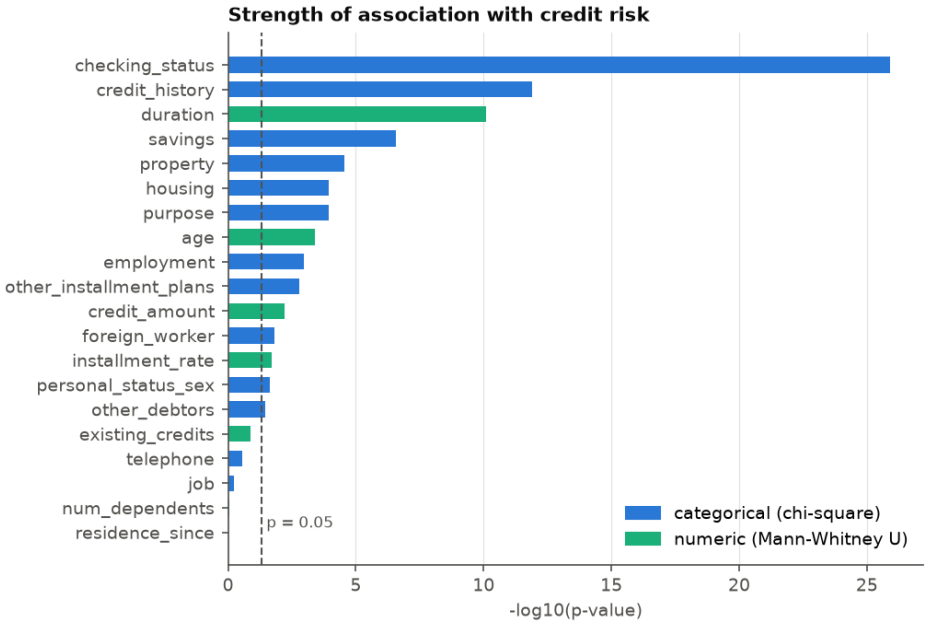


Figure 4. Strength of association (-log10 p) for every attribute.

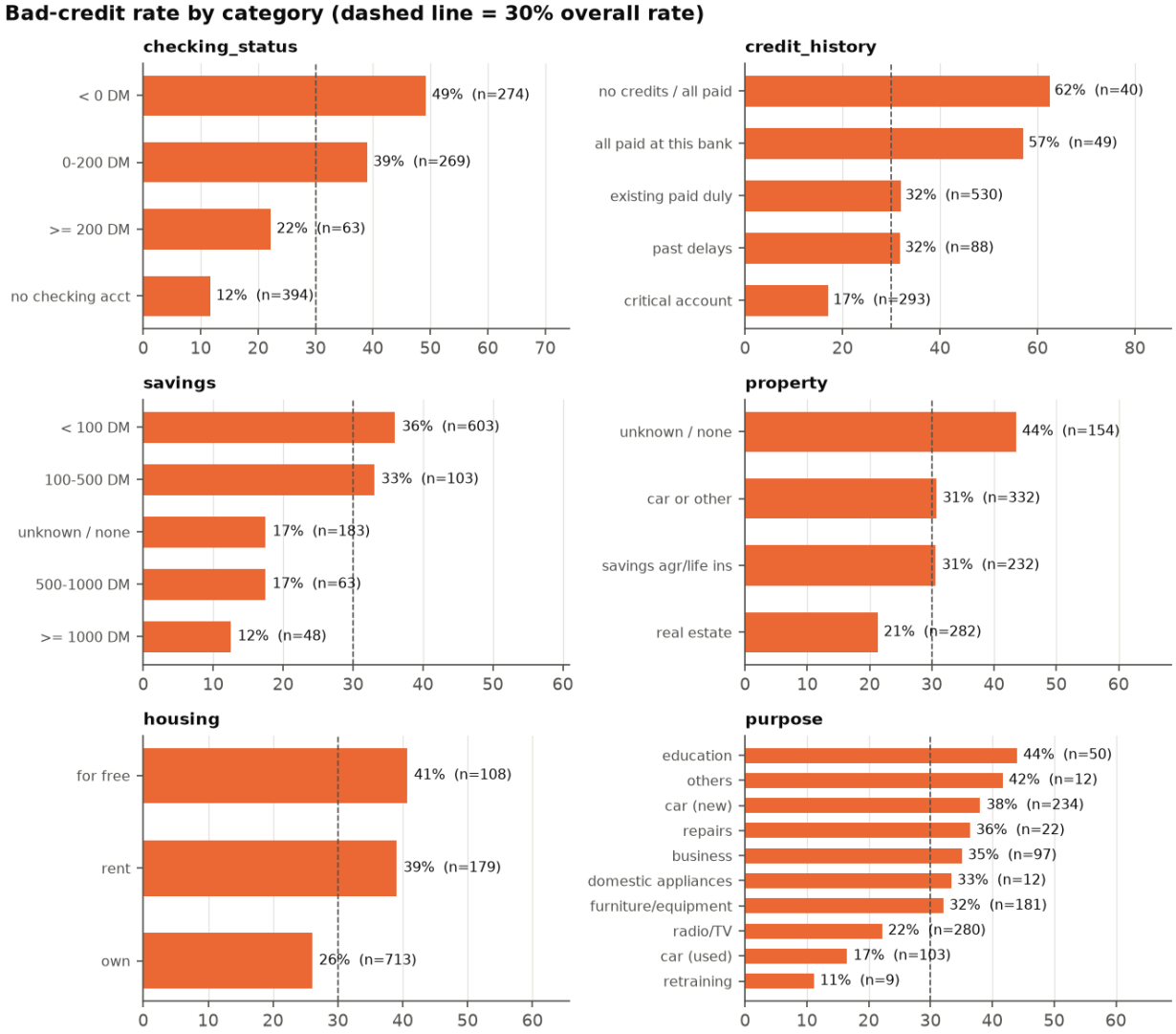


Figure 5. Bad-credit rate within each category of the six most associated categorical attributes.

Key observations

- **Checking account:** risk falls steadily from an overdrawn account (49% bad) to >= 200 DM (22%). It is lowest for applicants with no checking account at this bank (12%).
- **Credit history behaves counter-intuitively:** applicants coded "critical account / other credits elsewhere" have the lowest bad rate (17%). Those with "no credits / all paid back" have the highest (62%, n = 40). This probably reflects how the bank selected applicants in the past (the data contain only granted loans), so it should not be read as a causal effect.
- **Savings and property act as a buffer:** savings >= 1000 DM (12% bad) and real estate ownership (21%) are protective. No known property raises the bad rate to 44%.
- **Housing:** owners default less (26%) than renters (39%) or people housed for free (41%).
- **Purpose:** education (44%) and new cars (38%) are the riskiest large groups. Used cars (17%) and radio/TV (22%) are the safest.

5. Predictive modelling (classification)

Pre-processing. Numeric attributes were standardised (z-scores). Categorical attributes were one-hot encoded, giving 61 model inputs. All transformations were placed inside a scikit-learn Pipeline, so they are fitted only on training folds and do not leak information.

Evaluation design. (a) Stratified 10-fold cross-validation on all 1,000 records, using out-of-fold probabilities. (b) A stratified 70/30 hold-out split (700 train / 300 test). Metrics: accuracy, ROC-AUC, F1 and recall for the *bad* class, and the **total misclassification cost** = 5 x FN + 1 x FP, where FN is a bad customer predicted good and FP is a good customer predicted bad. Bayes decision theory says to predict *bad* when P(bad) > $c_{FP}/(c_{FP}+c_{FN})$ = 1/6. Both the default 0.5 threshold and the cost-optimal 1/6 threshold are reported.

Models: logistic regression; decision tree (depth 4, min leaf 20); random forest (500 trees); gradient boosting (200 stages, depth 3); SVM with RBF kernel; k-nearest neighbours (k = 15); Gaussian naive Bayes.

Model	Accuracy @0.5	ROC-AUC (sd)	F1 bad @0.5	Recall bad @0.5	Cost @0.5	Cost @1/6	Accuracy @1/6
Random Forest	0.767	0.797 (0.029)	0.492	0.377	981	534	0.538
SVM (RBF)	0.767	0.796 (0.040)	0.537	0.450	893	508	0.604
Logistic Regression	0.752	0.785 (0.039)	0.532	0.470	884	527	0.629
Gradient Boosting	0.756	0.783 (0.029)	0.531	0.460	892	560	0.616
k-NN (k=15)	0.732	0.746 (0.050)	0.356	0.247	1172	587	0.569
Naive Bayes	0.666	0.724 (0.053)	0.557	0.700	694	625	0.651
Decision Tree	0.699	0.709 (0.049)	0.402	0.337	1097	622	0.562
Baseline: accept all	0.700	0.500	0.000	0.000	1500		
Baseline: reject all	0.300	0.500		1.000	700		

Table: 10-fold cross-validation results on 1,000 applicants, sorted by ROC-AUC.

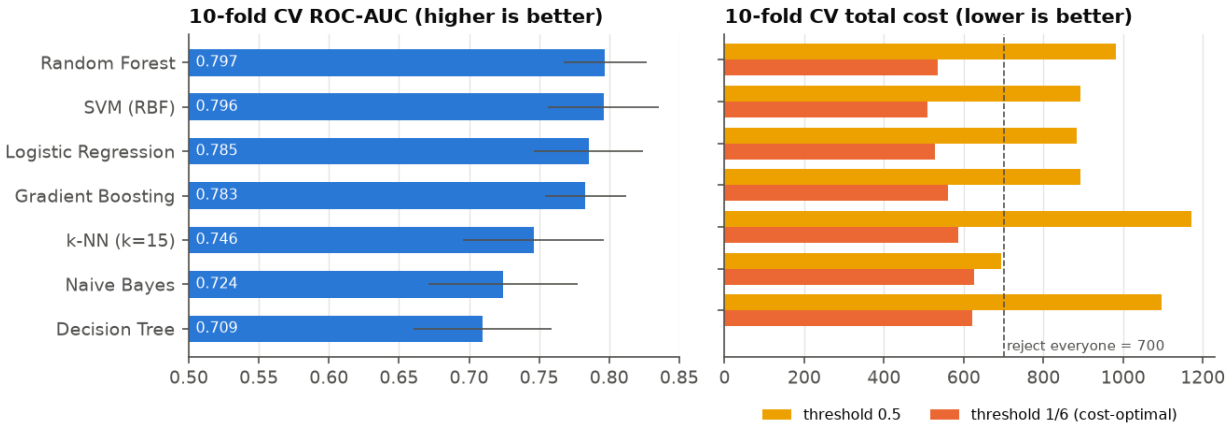


Figure 6. Left: cross-validated ROC-AUC (error bar = sd across folds). Right: total cost at the default and the cost-optimal threshold.

Hold-out test set (n = 300: 210 good, 90 bad)

Model	Thr	Accuracy	Prec. bad	Recall bad	F1 bad	AUC	TN	FP	FN	TP	Cost
Random Forest	0.5	0.763	0.686	0.389	0.496	0.800	194	16	55	35	291
Random Forest	1/6	0.547	0.395	0.956	0.558	0.800	78	132	4	86	152
SVM (RBF)	0.5	0.783	0.698	0.489	0.575	0.802	191	19	46	44	249
SVM (RBF)	1/6	0.533	0.386	0.944	0.548	0.802	75	135	5	85	160
Logistic Regression	0.5	0.783	0.676	0.533	0.596	0.801	187	23	42	48	233
Logistic Regression	1/6	0.633	0.444	0.878	0.590	0.801	111	99	11	79	154
Gradient Boosting	0.5	0.757	0.635	0.444	0.523	0.772	187	23	50	40	273
Gradient Boosting	1/6	0.653	0.457	0.822	0.587	0.772	122	88	16	74	168
k-NN (k=15)	0.5	0.743	0.667	0.289	0.403	0.766	197	13	64	26	333
k-NN (k=15)	1/6	0.563	0.397	0.878	0.547	0.766	90	120	11	79	175
Naive Bayes	0.5	0.697	0.496	0.678	0.573	0.726	148	62	29	61	207
Naive Bayes	1/6	0.670	0.469	0.744	0.575	0.726	134	76	23	67	191
Decision Tree	0.5	0.663	0.422	0.333	0.373	0.677	169	41	60	30	341
Decision Tree	1/6	0.567	0.397	0.856	0.542	0.677	93	117	13	77	182

Test-set baselines: accept everyone costs 450; reject everyone costs 210.

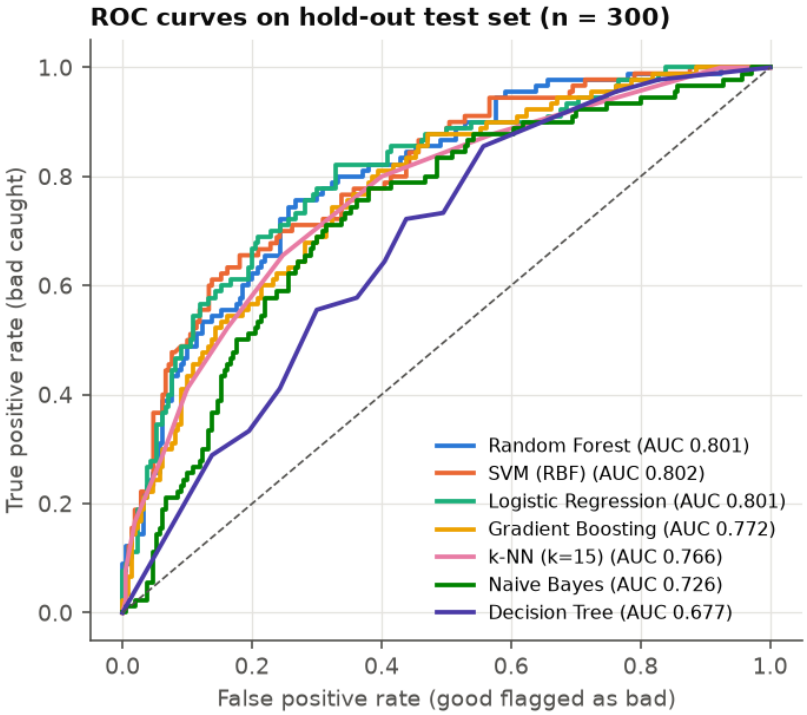


Figure 7. ROC curves on the hold-out test set.

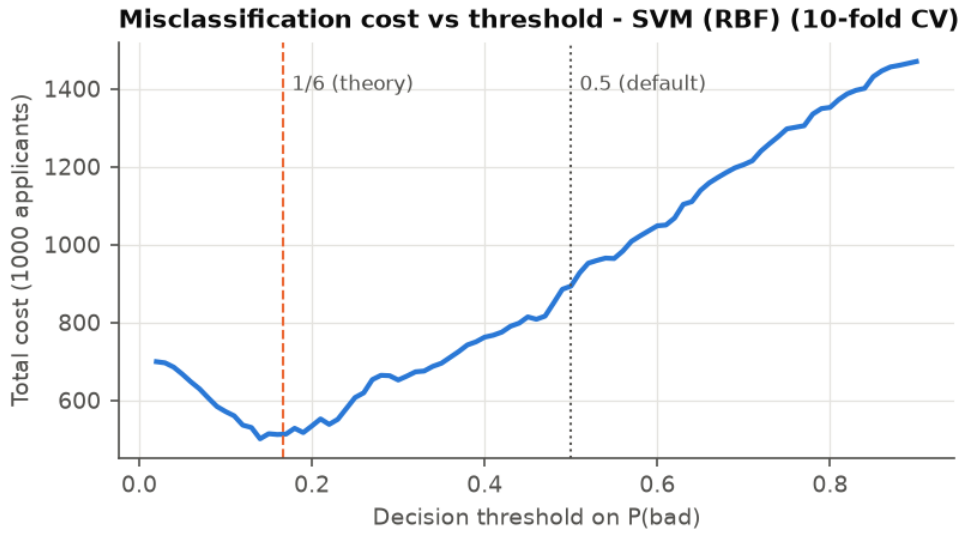


Figure 8. Cross-validated total cost of the SVM (RBF) model as a function of the threshold. The empirical minimum (cost 501 at threshold 0.14) lies close to the theoretical 1/6.

Interpretation of the modelling results

- Ensembles (random forest), the RBF-SVM and plain logistic regression perform equally well (CV AUC 0.785-0.797, overlapping fold-to-fold spreads). The simple, transparent logistic model gives up almost nothing in accuracy.
- The single decision tree, naive Bayes and k-NN are clearly weaker. One-hot encoded categorical data suits distance-based and independence-assuming learners poorly.
- At the default 0.5 threshold every model misses more than half of the bad customers (recall 0.25-0.47, except naive Bayes). Every model except naive Bayes then costs *more* than simply rejecting everyone (700).
- At the cost-optimal threshold of 1/6, recall for bad customers rises to 0.82-0.96 (naive Bayes 0.74). On the test set, logistic regression catches 79 of 90 bad applicants (cost 154) and random forest catches 86 of 90 (cost 152). That is less than 210 for rejecting everyone and 450 for accepting everyone. The price is that roughly 40-65% of good applicants are declined. The bank should decide explicitly whether the 5:1 cost ratio reflects its real economics.

6. Model interpretation: which features drive the predictions?

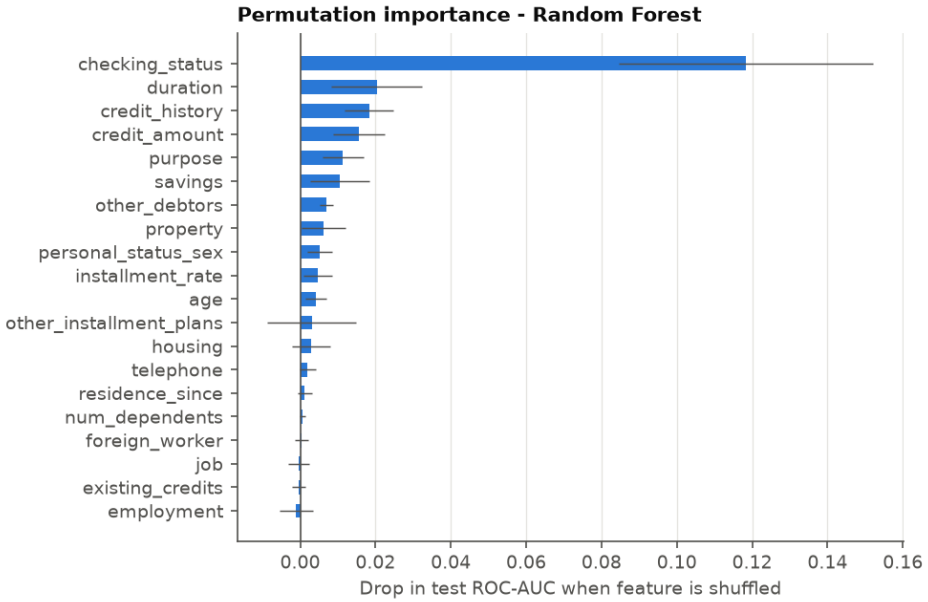


Figure 9. Permutation importance (random forest, hold-out test set, 20 repeats): the drop in ROC-AUC when one attribute is shuffled.

Checking-account status dominates the random forest (AUC drop 0.118). It is followed by duration, credit_history, credit_amount, purpose and savings. This agrees closely with the univariate tests in Section 4.

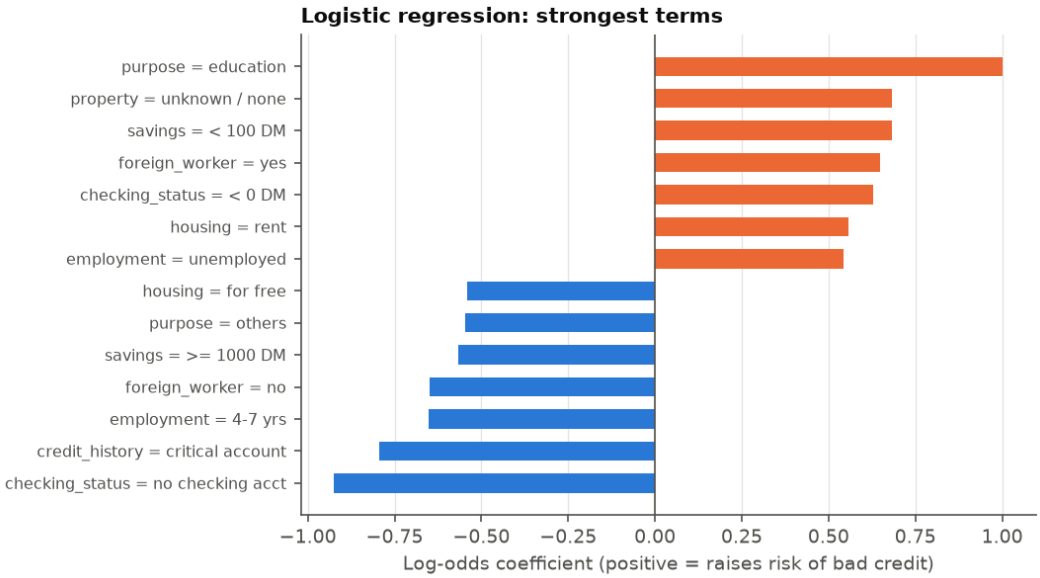


Figure 10. Largest logistic-regression coefficients (log-odds of being bad). Orange terms raise risk; blue terms lower it.

Term	Coefficient	Odds ratio
purpose = education	+1.001	2.72
checking_status = no checking acct	-0.924	0.40
credit_history = critical account	-0.795	0.45
property = unknown / none	+0.684	1.98
savings = < 100 DM	+0.683	1.98
employment = 4-7 yrs	-0.653	0.52
foreign_worker = no	-0.650	0.52
foreign_worker = yes	+0.648	1.91
checking_status = < 0 DM	+0.628	1.87
savings = >= 1000 DM	-0.567	0.57

Term	Coefficient	Odds ratio
housing = rent	+0.556	1.75
purpose = others	-0.545	0.58

Holding the other attributes constant, the odds of bad credit are about 2.7x higher for education loans. They are about 2x higher for applicants with no known property or savings under 100 DM, and about 1.9x higher with an overdrawn checking account. Having no checking account at this bank, a "critical" credit history (credits elsewhere), 4-7 years of employment or a guarantor all lower the odds.

The shallow decision tree (depth 4) makes its first split on checking-account status. It then splits on duration, savings and property, matching the variables identified above.

7. Customer segmentation (k-means clustering)

Unsupervised k-means was run on the same standardised, one-hot encoded data, without using the target. Silhouette scores for k = 2..7 were k=2: 0.121, k=3: 0.093, k=4: 0.099, k=5: 0.090, k=6: 0.081, k=7: 0.067. These low values (< 0.15) show there are **no strongly separated natural groups**: applicants form a continuum. k = 4 was kept because it gives segments that are easy to interpret and differ clearly in risk.

Cluster	Size	Bad rate	Median age	Median duration (m)	Median amount (DM)	Inst. rate	Typical purpose	Typical checking
0	264	19%	44	17	1,922	4	radio/TV	no checking acct
1	141	30%	38	15	2,442	3	car (new)	no checking acct
2	447	30%	27	15	1,950	3	radio/TV	no checking acct
3	148	50%	32	36	7,788	2	car (used)	0-200 DM

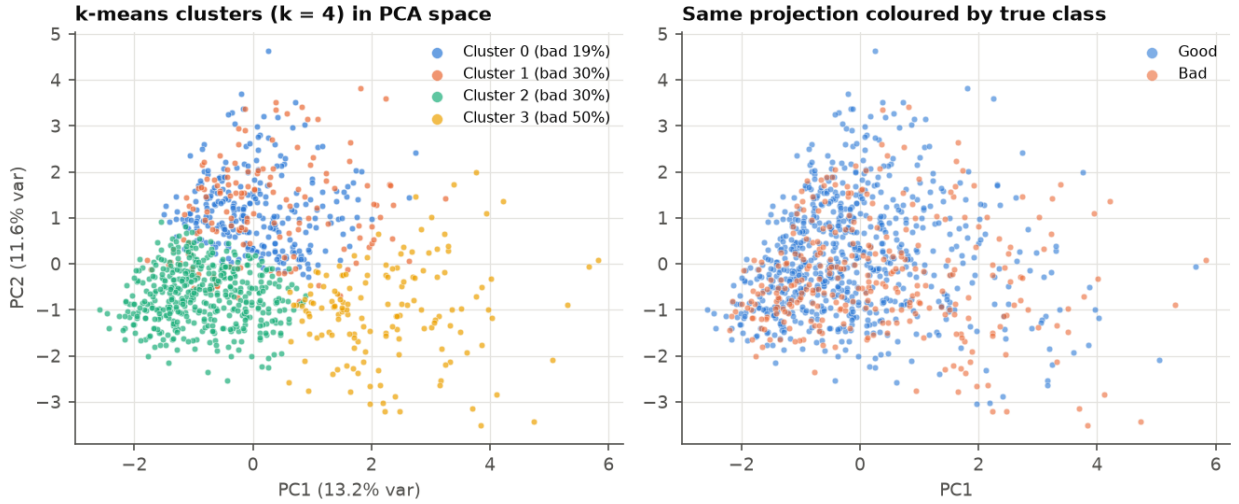


Figure 11. Clusters (left) and true classes (right) projected on the first two principal components (13.2% + 11.6% of variance).

- **Low risk segment (cluster 0, 19% bad):** older applicants (median 44) with several existing credits, small-to-medium loans and long residence.
- **High risk segment (cluster 3, 50% bad):** long (median 36 months), large (median 7,788 DM) loans. Half of these applicants turn out bad.
- The two middle clusters, one dominated by younger applicants, sit near the 30% average.
- In the right panel the good and bad classes overlap heavily. This explains why no classifier exceeds an AUC of about 0.80.

8. Association-rule mining (Apriori)

Numeric attributes were discretised (duration: <=12, 13-24, 25-36, >36 months; credit amount: terciles; age: <=25, 26-35, 36-50, >50). Eleven attributes plus the class label were then converted to a transaction table. Frequent itemsets were mined with Apriori (min support 3% = 30 applicants, up to 4 items). Rules were kept if their consequent is **class = BAD**, confidence >= 50% and lift >= 1.5 (46 rules). Redundant rules, whose condition extends a rule that is at least as strong, were removed.

IF (antecedent)	THEN	Support	Count	Confidence	Lift
duration=>36m AND savings=< 100 DM	BAD	0.034	34	0.72	2.41
checking_status=< 0 DM AND housing=rent AND savings=< 100 DM	BAD	0.033	33	0.64	2.12
checking_status=< 0 DM AND property=unknown / none	BAD	0.031	31	0.62	2.07
checking_status=< 0 DM AND property=car or other AND savings=< 100 DM	BAD	0.034	34	0.62	2.06
checking_status=< 0 DM AND credit_amount=low AND credit_history=existing paid duly	BAD	0.031	31	0.61	2.03
age=26-35 AND checking_status=< 0 DM AND savings=< 100 DM	BAD	0.044	44	0.60	2.01
checking_status=< 0 DM AND other_installment_plans=none AND property=car or other	BAD	0.032	32	0.59	1.98
age=<=25 AND checking_status=< 0 DM AND other_installment_plans=none	BAD	0.032	32	0.58	1.94
age=<=25 AND checking_status=< 0 DM AND savings=< 100 DM	BAD	0.030	30	0.58	1.92
credit_amount=high AND property=unknown / none AND savings=< 100 DM	BAD	0.030	30	0.58	1.92
checking_status=< 0 DM AND purpose=car (new) AND savings=< 100 DM	BAD	0.036	36	0.57	1.90
checking_status=< 0 DM AND credit_amount=high AND savings=< 100 DM	BAD	0.045	45	0.57	1.90

Strongest rules leading to GOOD credit (confidence >= 85%)

IF (antecedent)	THEN	Support	Count	Confidence	Lift
checking_status=no checking acct AND other_installment_plans=none AND purpose=car (used)	GOOD	0.046	46	1.00	1.43
checking_status=no checking acct AND duration=<=12m AND property=real estate	GOOD	0.060	60	0.98	1.41
checking_status=no checking acct AND credit_history=critical account AND purpose=radio/TV	GOOD	0.054	54	0.98	1.40
checking_status=no checking acct AND credit_amount=low AND property=real estate	GOOD	0.053	53	0.98	1.40
checking_status=no checking acct AND credit_amount=mid AND purpose=radio/TV	GOOD	0.050	50	0.98	1.40
age=26-35 AND checking_status=no checking acct AND property=real estate	GOOD	0.045	45	0.98	1.40
checking_status=no checking acct AND credit_history=critical account AND other_installment_plans=none	GOOD	0.131	131	0.98	1.40
checking_status=no checking acct AND property=real estate AND purpose=radio/TV	GOOD	0.038	38	0.97	1.39

The rules tell the same story in a form loan officers can read directly. Bad credit concentrates where an **overdrawn checking account or long duration** is combined with **low savings, renting or no property**; these groups default at roughly twice the base rate (lift about 2). Almost risk-free groups (97-100% good) combine **no checking account at this bank** with real estate, short loans, used-car or radio/TV purposes, or a critical credit history.

9. Sensitive attributes and fairness

The UCI variable table flags several demographic attributes: personal status and sex, age, job, employment, housing and foreign-worker status. Several of these are significantly associated with the target (age $p < 0.001$; foreign worker $p = 0.016$; personal status/sex $p = 0.022$). The share of applicants flagged as bad by the cost-optimal SVM (RBF) model (threshold 1/6, out-of-fold) differs by group:

Group	n	Actual bad rate	Flagged as bad by model
age <= 25	190	42.1%	81.1%
age > 25	810	27.2%	60.0%
foreign worker = yes	963	30.7%	64.8%
foreign worker = no	37	10.8%	43.2%
female (A92/A95)	310	35.2%	71.6%
male (A91/A93/A94)	690	27.7%	60.6%

Applicants aged 25 or under and women are flagged noticeably more often. Part of this follows from their higher observed bad rates, but the cost-sensitive threshold amplifies the gap. Only 37 applicants are non-foreign workers, so estimates for that group are unreliable. In most jurisdictions sex and nationality may not be used for credit decisions. A deployable model should drop or neutralise these attributes and be re-validated with fairness metrics (demographic parity, equal opportunity).

10. Conclusions, recommendations and limitations

Information gathered

- Data: 1,000 complete records with no missing or duplicate values. 20 attributes, 70% good / 30% bad.
- Most informative attributes: checking-account status, credit history, duration, savings, property, purpose and housing. Residence duration, number of dependents, job and telephone carry almost no signal.
- Risk profile of a bad applicant: overdrawn or low-balance checking account, long and large loan, little savings, no property, renting, often a new car or education purpose, and younger age.
- Best predictive performance: ROC-AUC of about 0.80 (random forest, SVM, logistic regression). The data overlap strongly between classes, which limits what any model can achieve.
- Using the cost matrix is essential. Lowering the threshold to P(bad) > 1/6 cuts the expected cost by roughly 40% compared with the default 0.5 rule.
- Segments: k-means finds a low-risk segment of older, established customers (19% bad) and a high-risk segment of large, long-term loans (50% bad).

Recommendations

- Use a **cost-sensitive logistic regression** or random forest scorecard with a threshold near 1/6. Logistic regression matches the ensembles and is easier to explain to regulators and customers.
- Add manual review, or require collateral or a guarantor, for the high-risk patterns found by association mining: overdrawn account with low savings, and loans over 36 months with savings under 100 DM.
- Remove protected attributes (sex, foreign-worker status) and monitor age effects before deployment.
- Re-estimate the 5:1 cost ratio from actual loss and profit data, because the threshold depends directly on it.

Limitations

- Small sample (1,000 records), collected in Germany in the 1970s-1990s in Deutsche Mark. The results may not transfer to current lending.
- Only approved loans are observed (reject-inference problem). This explains counter-intuitive effects such as the "critical account" history.
- The personal-status attribute mixes sex and marital status. One category (single female) has no records.
- The hyper-parameters are reasonable defaults rather than exhaustively tuned. Small AUC differences (< 0.01) between the top models are not significant.

Reproducibility

All code is in the repository: `src/load_data.py` (ucimlrepo loading), `src/analysis.py` (all computations and figures, random seed 42) and `src/build_report.py` (this PDF). Run `pip install -r requirements.txt`, then `python src/analysis.py && python src/build_report.py`.